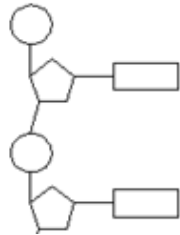


number			
3(a)	An answer that includes the following points: - two mononucleotides shown (joined in one strand only) (1) - each base bonded to C1 of pentose sugar (1) - phosphate group bonded to C3 of one sugar and C5 of the other sugar (1)	IGNORE labels including second strand if drawn / additional mononucleotides / one mononucleotide including second strand if drawn / additional mononucleotides <u>example of diagram</u> 	(3)

Question number	Answer	Additional guidance	Mark
3(b)(i)	0.38 (nm)	ACCEPT 0.4	(1)

number			
3(b)(ii)	An explanation that includes the following points: • R (or Q) because it forms two hydrogen bonds (1) • R (or S) because it is {large / double-ring / purine} base (1)	ACCEPT not bases P and S because they form 3 hydrogen bonds ACCEPT not bases P and Q because they are {small / single-ring / pyrimidine} ACCEPT Q for correct reason or S for correct reason if neither mark points awarded	(2)

number			
3(c)	An answer that includes at least one similarity and one difference: similarities: • both contain {RNA (mono)nucleotides / ribose sugar / uracil (and adenine, cytosine and guanine) / phosphodiester bonds} (1) • both are single stranded (1) differences: • mRNA is a straight chain and tRNA is {folded / clover-leaf shaped} / mRNA does not have hydrogen bonds and tRNA does (1) • {size / length} of mRNA is variable and the {size / length} of tRNA is constant (1) • mRNA has codons and tRNA has {anticodons / amino acid binding sites} (1)	DO NOT PIECE TOGETHER ACCEPT letters for bases	(4)